Problem No: 01

Problem Name: Study and Simulation of the 3D Radiation Pattern of a Dipole Antenna

Objective

- To understand the basic structure and working principle of a dipole antenna.
- To study the mathematical model of the dipole antenna radiation pattern.
- To simulate and visualize the 3D radiation pattern of a dipole antenna using Python.

Theory

A dipole antenna is one of the simplest and most widely used antennas in wireless communication systems. It consists of two conductive elements and is commonly used for transmitting and receiving electromagnetic waves. When the total length of the antenna is approximately half of the wavelength ($\lambda/2$), it is called a half-wave dipole antenna. It is widely used in radio communication and broadcasting because of its simple structure and reliable performance.

The radiation pattern of an antenna describes how electromagnetic energy is radiated in different directions. It shows the directions of maximum and minimum radiation. For simplified visualization, the normalized radiation pattern of a dipole antenna is expressed as:

$$R(\theta) = |\sin\theta|$$

Here, $R(\theta)$ represents the normalized radiation value, and θ is the angle measured from the antenna axis. The radiation is maximum at $\theta = 90^\circ$, where $R(\theta) = 1$. It becomes zero at $\theta = 0^\circ$ and $\theta = 180^\circ$. This means that the antenna radiates most strongly in the direction perpendicular to its axis and does not radiate along its axis in this model.

The radiation pattern can be represented in both two-dimensional (2D) and three-dimensional (3D) forms. The 2D pattern has a figure-eight shape in a suitable plane, while the 3D pattern has a doughnut-shaped structure, also called a toroidal pattern. The antenna radiates equally in all azimuth directions in this simplified model, but its radiation varies with the angle measured from the antenna axis.

For 3D visualization, the radiation pattern is first calculated using spherical coordinates and then converted into Cartesian coordinates. The transformation equations are:

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$$X = R \sin\theta \cos\phi$$

$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

Here, X, Y, and Z represent the Cartesian coordinates, and ϕ is the azimuth angle ranging from 0 to 2π. These equations help determine the position of each point on the 3D radiation surface.

Two important parameters of an antenna are directivity and radiation resistance. Directivity describes how strongly an antenna radiates in a particular direction compared with an ideal reference source. A half-wave dipole antenna has a typical directivity of approximately 2.15 dBi. Radiation resistance represents the radiated power in terms of an equivalent resistance.

In this experiment, Python is used to simulate and visualize the 3D radiation pattern. NumPy performs the mathematical calculations, while Matplotlib creates the 3D surface plot. The simulation helps us understand how radiation changes with direction and how the radiation pattern appears in three-dimensional space. Thus, the experiment demonstrates the basic radiation characteristics of a dipole antenna.

Source Code:

```
# Dipole Antenna - 3D Radiation Pattern

import numpy as np
import matplotlib.pyplot as plt

# Angle values
theta = np.linspace(0, np.pi, 100)
phi = np.linspace(0, 2 * np.pi, 100)

# Create coordinate grids
theta, phi = np.meshgrid(theta, phi)

# Simplified normalized radiation pattern
R = np.abs(np.sin(theta))

# Convert spherical coordinates to Cartesian coordinates
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)
```

```
# Create a 3D plot
fig = plt.figure()
ax = fig.add_subplot(projection='3d')

# Plot the radiation surface
ax.plot_surface(X, Y, Z)

# Add title and axis labels
ax.set_title("3D Radiation Pattern of Dipole Antenna")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")

# Display the plot
plt.show()
```

Result and discussion:

The 3D radiation pattern of a dipole antenna was successfully simulated and visualized using Python. The resulting pattern exhibits a doughnut-shaped structure, indicating maximum radiation perpendicular to the antenna axis and zero radiation along the axis. According to the simplified mathematical model, $R(\theta) = |\sin \theta|$, the normalized radiation is maximum at $\theta = 90°$ and zero at $\theta = 0°$ and $\theta = 180°$.

The pattern remains uniform in the azimuth direction because the radiation model depends only on the angle θ. These observations agree with the mathematical model used in the simulation. Therefore, the experiment successfully demonstrates the three-dimensional radiation characteristics of a dipole antenna.

Problem No: 02

Problem Name: Study and Simulation of the 3D Radiation Pattern of an Isotropic Antenna

Objective

- To understand the concept of an isotropic antenna.
- To study the uniform radiation characteristics of an isotropic antenna in all directions.
- To analyze the mathematical model of an ideal isotropic antenna.
- To simulate and visualize the 3D radiation pattern of an isotropic antenna using Python.

Theory — Isotropic Antenna

An isotropic antenna is an ideal and hypothetical antenna that radiates electromagnetic energy equally in all directions in three-dimensional space. Unlike a dipole antenna, it has no preferred direction of radiation. Although it cannot be physically realized, it is widely used as a reference model for comparing the performance of practical antennas.

The radiation pattern of an antenna describes how electromagnetic energy is radiated in different directions. It shows the directions of maximum and minimum radiation. For an isotropic antenna, the normalized radiation pattern is expressed as:

$$R(\theta, \phi) = 1$$

Here, $R(\theta, \phi)$ represents the normalized radiation value, θ is the polar angle, and ϕ is the azimuth angle. Since the radiation value is constant for all angles, the antenna radiates equally in every direction. Therefore, there are no radiation nulls or preferred directions.

The radiation pattern can be represented in both two-dimensional (2D) and three-dimensional (3D) forms. The 2D radiation pattern appears as a perfect circle, while the 3D radiation pattern forms a perfect sphere. This spherical shape indicates that radiation is uniform in all directions.

For 3D visualization, the radiation pattern is represented using spherical coordinates and converted into Cartesian coordinates. The transformation equations are:

$$X = R \sin\theta \cos\phi$$

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$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

Here, X, Y, and Z represent the Cartesian coordinates, and ϕ is the azimuth angle ranging from 0 to 2π. These equations help generate the 3D radiation surface. Since $R = 1$ in all directions, the resulting surface is a perfect sphere.

An isotropic antenna has a theoretical directivity of 1, equivalent to 0 dBi. Directivity describes how concentrated the radiation is in a particular direction. The isotropic antenna is used as a reference for expressing the gain of practical antennas in dBi.

In this experiment, Python is used to simulate and visualize the 3D radiation pattern. NumPy performs the mathematical calculations, while Matplotlib creates the 3D surface plot. The simulation helps us understand the spatial distribution and directional characteristics of an

isotropic antenna. Thus, the experiment demonstrates the uniform radiation characteristics of an ideal isotropic antenna.

Source Code:

```
# Isotropic Antenna - 3D Radiation Pattern

import numpy as np
import matplotlib.pyplot as plt

# Angle values
theta = np.linspace(0, np.pi, 200)
phi = np.linspace(0, 2 * np.pi, 200)

# Create coordinate grids
theta, phi = np.meshgrid(theta, phi)

# Constant normalized radiation pattern
R = np.ones_like(theta)

# Convert spherical coordinates to Cartesian coordinates
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

# Create a 3D plot
fig = plt.figure()
ax = fig.add_subplot(projection='3d')

# Plot the radiation surface
ax.plot_surface(X, Y, Z, cmap="plasma")

# Add title and axis labels
ax.set_title("3D Radiation Pattern of Isotropic Antenna")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")

# Display the plot
plt.show()
```

Result and Discussion

The 3D radiation pattern of an isotropic antenna was successfully simulated and visualized using Python. The resulting pattern forms a perfect sphere, showing that the normalized radiation value remains constant in all directions. The radiation value is set to 1 for every angle, so there are no radiation nulls or preferred directions. The spherical shape agrees with the theoretical model of an ideal isotropic antenna. Therefore, the experiment successfully demonstrates the uniform radiation characteristics of an isotropic antenna and its importance as a reference for comparing practical antennas.

Problem No: 03

Problem Name: Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective

- To understand the working principle of a horn antenna.
- To study the directional radiation characteristics of a horn antenna.
- To analyze the main beam, beamwidth, and directivity.
- To simulate and visualize the approximate 3D radiation pattern using Python.

Theory

A horn antenna is a highly directional antenna that operates efficiently in the microwave frequency range. It is formed by gradually flaring the end of a waveguide, which improves radiation into free space and helps provide good impedance matching.

Unlike an isotropic antenna, which radiates equally in all directions, a horn antenna concentrates electromagnetic energy mainly in the forward direction. Therefore, it provides high directivity and gain. Its radiation pattern consists of a main lobe and may also contain smaller side lobes.

In this experiment, an approximate mathematical model is used to visualize the radiation characteristics of a horn antenna. The aperture effect is represented using sinc functions, and backward radiation is reduced using a cosine-based factor.

The approximate radiation pattern is expressed as:

$$E(\theta, \phi) = \left| \text{sinc}\left(\frac{u}{\pi}\right) \text{sinc}\left(\frac{v}{\pi}\right) \right| \frac{1 + \cos\theta}{2}$$

where,

$$u = 10 \sin\theta \cos\phi$$

$$v = 8 \sin\theta \sin\phi$$

Here, θ represents the polar angle and ϕ represents the azimuth angle. The coefficients 10 and 8 are illustrative parameters used in the simulation to control the pattern shape.

The radiation pattern is normalized using:

$$E_{\text{norm}} = \frac{E}{E_{\max}}$$

The normalized radiation is then converted into decibel scale using:

$$E_{\text{dB}} = 20 \log_{10}(E_{\text{norm}})$$

For visualization, the radiation level is limited to -40 dB and scaled to a range between 0 and 1.

The radiation pattern is converted from spherical coordinates to Cartesian coordinates using:

$$X = E \sin\theta \cos\phi$$

$$Y = E \sin\theta \sin\phi$$

$$Z = E \cos\theta$$

Finally, Python and Matplotlib are used to plot the approximate 3D radiation pattern. This visualization helps illustrate the directional behavior of a horn antenna. However, it is a simplified mathematical model rather than an exact electromagnetic simulation of a practical horn antenna.

Source Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Define angles
theta = np.linspace(0, np.pi, 220)
phi = np.linspace(0, 2*np.pi, 240)

TH, PH = np.meshgrid(theta, phi)

# Approximate aperture effect
u = 10*np.sin(TH)*np.cos(PH)
v = 8*np.sin(TH)*np.sin(PH)

# Calculate approximate radiation pattern
E = np.abs(np.sinc(u/np.pi) * np.sinc(v/np.pi)) \
    * (1 + np.cos(TH))/2

# Normalize radiation values
E = E / E.max()

# Convert to dB scale and limit to -40 dB
E = np.maximum(
    20*np.log10(np.maximum(E, 1e-6)), -40
)

# Normalize values to the range 0 to 1
E = (E + 40) / 40

# Convert to Cartesian coordinates
X = E*np.sin(TH)*np.cos(PH)
Y = E*np.sin(TH)*np.sin(PH)
Z = E*np.cos(TH)

# Plot the 3D radiation pattern
fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")

ax.plot_surface(X, Y, Z, cmap="inferno", linewidth=0)

ax.set_title("Approximate 3D Horn Antenna Pattern")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
```

```
ax.set_box_aspect((1, 1, 1))
ax.set_zlim(-0.3, 1)

plt.tight_layout()
plt.show()
```

Result and Discussion :

The approximate 3D radiation pattern of a horn antenna was successfully simulated using Python. The graph illustrates the directional radiation characteristics of the antenna, with stronger radiation in the forward direction and reduced backward radiation. The sinc functions represent the approximate aperture effect, while the cosine-based factor reduces backward radiation. The 3D visualization helps demonstrate how a horn antenna concentrates electromagnetic energy in a particular direction. However, the simulation is based on a simplified mathematical model and does not represent the exact radiation pattern of a practical horn antenna.

Problem No: 03

Problem Name: Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective

- To understand the working principle of a horn antenna.
- To study the directional radiation characteristics of a horn antenna.
- To analyze the main lobe and directivity of the radiation pattern.
- To simulate and visualize the 3D radiation pattern using Python.

Theory

A horn antenna is a directional antenna that operates efficiently in the microwave frequency range. It is formed by gradually flaring the end of a waveguide, which helps electromagnetic waves radiate efficiently into free space. This design also helps improve impedance matching between the waveguide and free space.

Unlike an isotropic antenna, which radiates equally in all directions, a horn antenna focuses electromagnetic energy mainly in a specific direction. As a result, it provides high directivity and gain. The radiation pattern of a horn antenna is characterized by a dominant main lobe in the forward direction, indicating that most of the radiated energy is concentrated along a particular axis.

In this experiment, the radiation pattern is approximated using a cosine-based function:

$$R(\theta) = \cos^n(\theta)$$

where n is a constant that controls the beamwidth of the radiation pattern. A higher value of \(n\) produces a narrower main lobe, indicating stronger beam focusing. In this simulation, \(n=6\) is used, and the angular range is limited to the forward hemisphere:

$$0 \leq \theta \leq \frac{\pi}{2}$$

At $\theta = 0°$, the radiation reaches its maximum value, while it decreases as the angle approaches $90°$. This model represents a single forward lobe and does not include the backward hemisphere.

For three-dimensional visualization, the radiation pattern is converted from spherical coordinates to Cartesian coordinates using the following relations:

$$X = R \sin\theta \cos\phi$$

$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

where θ is the polar angle and ϕ is the azimuth angle.

The resulting 3D radiation pattern shows a concentrated beam along the positive Z-axis. This directional property makes horn antennas suitable for applications such as radar systems, satellite communication, microwave links, and antenna measurements. Horn antennas are also widely used as standard gain antennas because of their predictable radiation characteristics.

Thus, this experiment provides an understanding of the directional radiation characteristics of a horn antenna through a simplified mathematical model.

```
# Horn Antenna - 3D
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi/2, 250)    # শুধু সামনের hemisphere
phi = np.linspace(0, 2*np.pi, 250)
theta, phi = np.meshgrid(theta, phi)

n = 6
```

```
R = np.cos(theta)**n                # এখানে abs লাগে না, cos ≥ 0

X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="inferno")
ax.set_xlim(-1, 1)
ax.set_ylim(-1, 1)
ax.set_zlim(0, 1)
ax.set_title("Horn Antenna (3D)")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
plt.show()
```

Result and Discussion

The 3D radiation pattern of a horn antenna was successfully simulated using Python. The graph shows a dominant forward lobe along the positive Z-axis, indicating that the radiation is concentrated mainly in one direction. The radiation strength is maximum at $\theta = 0°$ and decreases as the angle approaches $90°$. The value $n = 6$ produces a relatively narrow beam, demonstrating the directional characteristics of the antenna. These characteristics make horn antennas suitable for radar systems, satellite communication, and microwave transmission. The simulation provides a simplified visualization of horn antenna radiation, although it does not represent the exact electromagnetic radiation pattern of a practical horn antenna.

Problem No: 04

Problem Name: Study and Simulation of the 2D Radiation Pattern of a Linear Antenna Array

Objectives

- To understand the working principle of a linear antenna array.
- To study the effect of multiple antenna elements on the radiation pattern.
- To observe the main lobes and side lobes of the array.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A linear antenna array consists of two or more identical antenna elements arranged in a straight line. These antenna elements work together to transmit or receive electromagnetic waves. The radiation pattern of the array depends on the number of elements, the spacing between adjacent elements, the amplitude, and the phase of the signals supplied to the elements.

The radiation pattern of an antenna array is determined by the individual element pattern and the array factor. The array factor (AF) represents the combined effect of the signals radiated by all the antenna elements. It helps to determine the directions of maximum and minimum radiation.

The array factor of a uniform linear antenna array is expressed as,

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jn(kd\cos\theta+\beta)}$$

Where,

- N = Number of antenna elements
- d = Spacing between adjacent elements
- $k = 2\pi/\lambda$ = Wave number
- λ = Wavelength
- β = Progressive phase shift
- θ = Observation angle

The main lobe is the part of the radiation pattern in the direction of maximum radiation. The side lobes are smaller lobes that appear in other directions. The beamwidth describes the angular width of the main lobe. Generally, increasing the number of antenna elements can produce a narrower main beam and improve directivity.

The spacing between antenna elements also affects the radiation pattern. A common spacing is half the wavelength ($d = \lambda/2$), which helps avoid grating lobes under typical uniform-array conditions.

When the progressive phase shift is zero ($\beta = 0$), the array produces broadside radiation, meaning that maximum radiation occurs perpendicular to the array axis. In this experiment, eight antenna elements are considered with half-wavelength spacing and zero phase shift. The array factor is calculated for different observation angles and normalized by its maximum value.

Finally, Python and Matplotlib are used to plot the normalized array factor on a polar graph. The graph helps visualize the main lobes, side lobes, and directional radiation characteristics of the linear antenna array.

Source Code:

```
# Linear Antenna Array - 2D Radiation Pattern
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
N = 8               # Number of elements
d = 0.5             # Spacing (lambda / 2)
k = 2 * np.pi       # Wave number
beta = 0            # Phase difference

# Angle
theta = np.linspace(0, 2 * np.pi, 1000)

# Array Factor
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * (k * d * np.cos(theta) + beta))

# Magnitude
AF = np.abs(AF)

# Normalize
AF = AF / np.max(AF)

# Plot
ax = plt.subplot(111, projection="polar")
ax.plot(theta, AF)

ax.set_title("Linear Antenna Array (2D)")

plt.show()
```

Result and Discussion

The 2D radiation pattern of a linear antenna array was successfully simulated using Python. The polar plot shows two main lobes in opposite broadside directions at approximately 90° and 270°. These directions represent maximum radiation perpendicular to the array axis.

The pattern also shows smaller side lobes, which represent radiation in other directions. The use of eight antenna elements with half-wavelength spacing and zero progressive phase shift demonstrates the directional characteristics of a linear antenna array. The obtained pattern is consistent with the expected behavior of a uniformly spaced array.

Problem No: 05

Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 3-Element Array Antenna

Objectives

- To understand the concept of a broadside array antenna.
- To study the radiation characteristics of a 3-element antenna array.
- To analyze the formation of the main lobe and radiation nulls.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna is a type of linear antenna array in which multiple identical antenna elements are arranged in a straight line with equal spacing. These antenna elements work together to transmit or receive electromagnetic waves. The radiation pattern of the array depends on the number of elements, the spacing between adjacent elements, the amplitude, and the phase of the signals supplied to the elements.

The radiation pattern of an antenna array is determined by the individual element pattern and the array factor. The array factor (AF) represents the combined effect of the signals radiated by all antenna elements. It helps to determine the directions of maximum and minimum radiation.

The array factor of a uniform linear antenna array is expressed as,

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jn(kd\cos\theta+\beta)}$$

Where,

- N = Number of antenna elements
- d = Spacing between adjacent elements
- $k = 2\pi/\lambda$ = Wave number
- λ = Wavelength
- β = Progressive phase shift
- θ = Observation angle

The main lobe is the part of the radiation pattern in the direction of maximum radiation. Radiation nulls are the directions where the array factor becomes zero. The beamwidth describes

the angular width of the main lobe. Generally, increasing the number of antenna elements can produce a narrower main beam and improve directivity.

The spacing between antenna elements also affects the radiation pattern. A common spacing is half the wavelength ($d = \lambda/2$), which helps avoid grating lobes under typical uniform-array conditions.

For a broadside array, the progressive phase shift is zero ($\beta = 0$), meaning all elements are excited with the same phase. This produces maximum radiation perpendicular to the array axis. In this experiment, three antenna elements are used with half-wavelength spacing and zero progressive phase shift.

The array factor is calculated for different observation angles and normalized by its maximum value. Normalization makes the maximum magnitude equal to 1, allowing the relative radiation strength to be observed clearly. Python and Matplotlib are used to plot the normalized array factor on a polar graph.

The 2D radiation pattern helps to visualize the main lobe, radiation nulls, and directional characteristics of the broadside 3-element array. This type of antenna array is useful in applications such as radar systems, wireless communication, and broadcasting, where directional radiation is required.

```
# Broadside Array Antenna (3-Element)

import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 3               # Number of elements

# Angle range
theta = np.linspace(0, np.pi, 1000)

# Array Factor
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6, 6))
```

```
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)

ax.set_title("2D Radiation Pattern (3-Element Broadside Array)")

plt.show()
```

Result and Discussion

The simulation results show the 2D radiation pattern of a 3-element broadside array antenna. The maximum radiation occurs in the broadside direction ($\theta = 90°$).

The main lobe is relatively broad compared to arrays with more elements. The radiation pattern demonstrates that equal-amplitude and in-phase excitation produces maximum radiation in the broadside direction. The results also show that the number of elements affects the beamwidth and overall radiation pattern.

The results are consistent with the theoretical behavior of a uniform broadside array and demonstrate the basic principle of directional radiation in antenna arrays.

Problem No: 06

Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Phase (\(d=\lambda/2\))

Objectives

- To understand the concept of interference between two isotropic sources.
- To study the far-field radiation pattern of a two-element antenna system.
- To analyze the effect of equal amplitude and phase on the radiation pattern.
- To simulate and visualize the radiation pattern using Python.

Theory

In antenna theory, when two isotropic point sources radiate electromagnetic waves simultaneously, the resulting radiation pattern is determined by the principle of superposition. An isotropic source is an ideal source that radiates equally in all directions. When two sources have equal amplitude and are in phase, their waves interfere constructively and destructively at different angles, producing a characteristic radiation pattern.

For two isotropic sources separated by a distance \(d\), the array factor (AF) in the far-field region is given by,

$$AF(\theta) = 2\cos\left(\frac{kd}{2}\cos\theta\right)$$

Where,

- $k = 2\pi/\lambda$ = Wave number
- λ = Wavelength
- d = Distance between the two sources
- θ = Observation angle

When the spacing is $d = \lambda/2$, the expression becomes,

$$AF(\theta) = 2\cos\left(\frac{\pi}{2}\cos\theta\right)$$

This equation shows that the radiation pattern depends on the observation angle and the interference between the waves from the two sources.

Constructive interference occurs when the waves combine to produce maximum radiation. Destructive interference occurs when the waves cancel each other, producing radiation nulls. The maximum radiation occurs in the broadside directions, perpendicular to the line joining the two sources.

In this experiment, two isotropic point sources are considered with equal amplitude, zero phase difference, and half-wavelength spacing. The array factor is calculated for different observation angles and normalized by its maximum magnitude. Normalization makes the maximum value equal to 1 and helps to visualize the relative radiation strength.

Python and Matplotlib are used to plot the normalized far-field radiation pattern on a polar graph. The pattern shows two main lobes in opposite broadside directions and nulls along the source-array axis. This experiment demonstrates how interference between multiple sources affects the radiation characteristics of an antenna system.

Source Code:

```
# Two Isotropic Point Sources (Equal Amplitude & Phase)
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam

# Angle range
theta = np.linspace(0, 2*np.pi, 1000)

# Array Factor
AF = 2 * np.cos((k * d / 2) * np.cos(theta))

# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6, 6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nEqual Amplitude & Phase (d = λ/2)")
plt.show()
```

Result and Discussion

The far-field radiation pattern of two isotropic point sources was successfully simulated using Python. The polar plot shows two main lobes in opposite broadside directions at approximately 90° and 270°. The radiation becomes zero at 0° and 180°, producing nulls along the source-array axis.

The pattern demonstrates the effects of constructive and destructive interference between the two sources. Equal amplitude, zero phase difference, and half-wavelength spacing produce the expected symmetric radiation pattern. The result agrees with the theoretical behavior of two isotropic point sources and illustrates how interference affects the radiation characteristics of an antenna system.

Problem No: 07

Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Opposite Phase (\(d=\lambda/2\))

Objectives

- To understand the effect of phase difference between two isotropic sources.

- To study the far-field radiation pattern for opposite-phase excitation.
- To analyze constructive and destructive interference.
- To simulate and visualize the radiation pattern using Python.

Theory

In antenna theory, when two isotropic point sources radiate electromagnetic waves simultaneously, the resulting radiation pattern is determined by the principle of superposition. An isotropic source is an ideal source that radiates equally in all directions. When two sources have equal amplitude but opposite phase, their waves interfere constructively and destructively at different angles, producing a characteristic radiation pattern.

For two isotropic sources separated by a distance d , the array factor (AF) considering a phase difference β is given by,

$$AF(\theta) = 2\cos\left(\frac{kd}{2}\cos\theta + \frac{\beta}{2}\right)$$

Where,

- $k = 2\pi/\lambda$ = Wave number
- λ = Wavelength
- d = Distance between the two sources
- β = Phase difference between the sources
- θ = Observation angle

For opposite-phase excitation, the phase difference is $\beta = \pi$, which means the sources have a phase difference of $180°$. Using a trigonometric identity, the array factor can be expressed as,

$$AF(\theta) = -2\sin\left(\frac{kd}{2}\cos\theta\right)$$

The negative sign does not affect the magnitude of the radiation pattern. Therefore, the magnitude can be calculated using,

$$|AF(\theta)| = 2\left|\sin\left(\frac{kd}{2}\cos\theta\right)\right|$$

For spacing $d = \lambda/2$, the expression becomes,

$$AF(\theta) = -2\sin\left(\frac{\pi}{2}\cos\theta\right)$$

This equation shows that the radiation pattern depends on the observation angle and the interference between the waves from the two sources.

Constructive interference occurs when the waves combine to produce maximum radiation, while **destructive interference** occurs when the waves cancel each other, producing radiation nulls. For opposite-phase excitation with half-wavelength spacing, maximum radiation occurs along the source-array axis at 0° and 180°. Nulls occur in the broadside directions at 90° and 270°.

In this experiment, two isotropic point sources are considered with equal amplitude, opposite phase, and half-wavelength spacing. The array factor is calculated for different observation angles and normalized by its maximum magnitude. Python and Matplotlib are used to plot the normalized far-field radiation pattern on a polar graph.

The simulation helps to understand how phase difference affects the direction of maximum radiation and changes the overall radiation pattern. This concept is important in antenna array design, where phase control is used to obtain desired radiation characteristics.

```
# Two Isotropic Point Sources (Opposite Phase)

import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam

# Angle range
theta = np.linspace(0, 2*np.pi, 1000)

# Array Factor (Opposite Phase)
AF = 2 * np.sin((k * d / 2) * np.cos(theta))
```

```
# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6, 6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Sources (Opposite Phase)\n(d = λ/2)")
plt.show()
```

Result and Discussion

The far-field radiation pattern of two isotropic point sources with equal amplitude and opposite phase was successfully simulated using Python. The polar plot shows two main lobes in opposite end-fire directions at approximately 0° and 180°. The radiation becomes zero at 90° and 270°, producing nulls in the broadside directions.

The pattern demonstrates the effects of constructive and destructive interference between the two sources. Opposite-phase excitation and half-wavelength spacing produce the expected end-fire type radiation pattern. The result agrees with the theoretical behavior of two isotropic point sources and illustrates how phase difference can control the direction of radiation.

Problem No: 08

Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Unequal Amplitude and Opposite Phase (d = λ/2)

Objective

- To understand the effect of unequal amplitude in antenna arrays.
- To study the radiation pattern for opposite phase excitation.
- To analyze how amplitude variation affects nulls in the radiation pattern.
- To simulate and visualize the radiation pattern using Python.

Theory

When two isotropic point sources radiate electromagnetic waves with unequal amplitudes and opposite phase, the resulting radiation pattern depends on the amplitude and phase difference of the sources. According to the principle of superposition, the total radiated field is the sum of the fields produced by the individual sources.

For two isotropic sources with amplitudes A_1 and A_2, separated by a distance d, the array factor (AF) is given by:

$$AF(\theta) = A_1 + A_2 e^{j(kd\cos\theta + \beta)}$$

where $k = 2\pi/\lambda$ is the wave number, λ is the wavelength, d is the spacing between the sources, and β is the phase difference between the sources.

For opposite phase excitation, the phase difference is $\beta = \pi$. Therefore, the array factor becomes:

$$AF(\theta) = A_1 - A_2 e^{jkd\cos\theta}$$

$$AF(\theta) = A_1 - A_2 e^{jkd\cos\theta}$$

This indicates that the signals from the two sources are out of phase, leading to destructive interference in certain directions and constructive interference in others.

When the amplitudes are unequal ($A_1 \neq A_2$), the signals cannot completely cancel each other. As a result, the nulls are no longer completely zero, unlike the equal amplitude case. The radiation pattern depends on the amplitude ratio of the two sources.

For a spacing of $d = \lambda/2$, the radiation pattern has maximum radiation at $\theta = 0°$ and 180°, while minimum radiation occurs at $\theta = 90°$ and 270°. In this experiment, the amplitudes are $A_1 = 1$ and $A_2 = 0.5$. Therefore, the minimum radiation is not zero, and the normalized minimum magnitude is approximately 0.333.

Thus, the experiment demonstrates how unequal amplitude affects the radiation pattern and the depth of nulls in an antenna array.

```
# Two Isotropic Sources (Unequal Amplitude + Opposite Phase)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam

# Amplitude weights
A1 = 1
A2 = 0.5  # Unequal amplitude
```

```
# Angle range
theta = np.linspace(0, 2*np.pi, 1000)

# Array Factor
AF = A1 - A2 * np.exp(1j * k * d * np.cos(theta))

# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nUnequal Amplitude + Opposite Phase\n(d = λ/2)")
plt.show()
```

Result and Discussion

The simulation shows the far-field radiation pattern of two isotropic point sources with unequal amplitudes and opposite phase. The amplitudes are $A_1 = 1$ and $A_2 = 0.5$, and the distance between the sources is $\lambda/2$.

The maximum radiation occurs at 0° and 180°, while the minimum radiation occurs at 90° and 270°. Unlike the equal-amplitude case, the minimum radiation is not zero because the two sources cannot completely cancel each other. The normalized minimum magnitude is approximately 0.333.

The pattern remains symmetric, but its nulls are not completely zero. The results agree with theoretical expectations and demonstrate the effect of unequal amplitude on the radiation pattern of an antenna array.

Problem No: 09

Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 5-Element Array Antenna

Objective

- To understand the behavior of a multi-element broadside array antenna.
- To study the effect of increasing the number of elements on the radiation pattern.
- To analyze the beamwidth and side lobe formation.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna consists of multiple identical elements arranged linearly with equal spacing and excited with equal amplitude and phase. In such an array, the maximum radiation occurs perpendicular to the array axis, known as the broadside direction (θ = 90°).

For a uniform linear array with N elements and spacing d, the array factor (AF) is expressed as:

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jnkd\cos\theta}$$

where k = 2π/λ is the wave number, λ is the wavelength, d is the spacing between elements, and N is the number of elements.

In a broadside configuration, all elements are excited in phase (β = 0), ensuring constructive interference in the perpendicular direction.

In this experiment, a 5-element array is used with a spacing of d = λ/2. Increasing the number of elements makes the main lobe narrower, indicating higher directivity and better beam focusing. Small side lobes also appear on both sides of the main beam, representing radiation in other directions.

Thus, increasing the number of elements improves the directional performance of an antenna array. This experiment demonstrates how the number of elements affects the radiation pattern, beamwidth, and side lobe formation.

```
# Broadside array antenna (5-element)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 5  # number of elements

# Angle range
theta = np.linspace(0, np.pi, 1000)

# Array Factor
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))
```

```
# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
ax.set_title("2D Radiation Pattern (5-Element Broadside Array)")
plt.show()
```

Result and Discussion

The simulation results show the 2D radiation pattern of a 5-element broadside array antenna. The maximum radiation occurs in the broadside direction (θ = 90°), while nulls occur at θ = 0° and 180°.

Compared to the 3-element array, the main lobe becomes narrower, indicating higher directivity and better beam focusing. Small side lobes also appear on both sides of the main beam. This demonstrates that increasing the number of elements improves directional performance and affects the shape of the radiation pattern.

The results are consistent with theoretical expectations and show that antenna arrays can control radiation patterns for applications such as radar and wireless communication systems.

Problem No: 10

Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 7-Element Array Antenna

Objective

- To study the behavior of a large antenna array system.
- To analyze the effect of increasing the number of elements on the radiation pattern.
- To observe beam narrowing and side lobe formation.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna consists of multiple identical elements arranged linearly with equal spacing and excited with equal amplitude and phase. In such an array, the maximum radiation occurs perpendicular to the array axis, known as the broadside direction (θ = 90°).

For a uniform linear array with N elements and spacing d, the array factor (AF) is expressed as:

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jnkd\cos\theta}$$

where k = 2π/λ is the wave number, λ is the wavelength, d is the spacing between elements, and N is the number of elements.

In a broadside configuration, all elements are excited in phase (β = 0), ensuring constructive interference in the perpendicular direction.

In this experiment, a 7-element array is used with a spacing of d = λ/2. As the number of elements increases, the main lobe becomes narrower, indicating higher directivity and better beam focusing. Multiple side lobes also appear on both sides of the main beam, representing radiation in other directions.

Thus, increasing the number of elements improves the directional performance of an antenna array but also increases the complexity of the radiation pattern. This experiment demonstrates the effect of the number of elements on beamwidth, directivity, and side lobe formation.

```
# Broadside array antenna (7-element)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 7  # number of elements

# Angle range
theta = np.linspace(0, np.pi, 1000)

# Array Factor
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
```

```
ax.set_title("2D Radiation Pattern (7-Element Broadside Array)")
plt.show()
```

Result and Discussion

The simulation results demonstrate the 2D radiation pattern of a 7-element broadside array antenna. The maximum radiation occurs in the broadside direction (θ = 90°), while nulls occur at θ = 0° and 180°.

Compared to arrays with fewer elements, the main lobe becomes narrower, indicating higher directivity and better beam focusing. Multiple side lobes are also observed on both sides of the main beam. This shows that increasing the number of elements improves directional performance but also introduces additional side lobes.

The results are consistent with theoretical expectations and demonstrate how antenna arrays can control radiation patterns for applications such as radar and wireless communication systems.

Problem No: 03

Problem Name: Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective

- To understand the working principle of a horn antenna.
- To study the directional radiation characteristics of a horn antenna.
- To analyze the main lobe and directivity of the radiation pattern.
- To simulate and visualize the 3D radiation pattern using Python.

Theory

A horn antenna is a directional antenna that operates efficiently in the microwave frequency range. It is formed by gradually flaring the end of a waveguide, which helps electromagnetic waves radiate efficiently into free space. This design also helps improve impedance matching between the waveguide and free space.

Unlike an isotropic antenna, which radiates equally in all directions, a horn antenna focuses electromagnetic energy mainly in a specific direction. As a result, it provides high directivity and gain. The radiation pattern of a horn antenna is characterized by a dominant main lobe in the forward direction, indicating that most of the radiated energy is concentrated along a particular axis.

In this experiment, the radiation pattern is approximated using a cosine-based function:

\[R(\theta)=\cos^n(\theta) \]

where n is a constant that controls the beamwidth of the radiation pattern. A higher value of n produces a narrower main lobe, indicating stronger beam focusing. In this simulation, $n=6$ is used, and the angular range is limited to the forward hemisphere:

$$ 0\leq\theta\leq\frac{\pi}{2} $$

At $\theta=0^\circ$, the radiation reaches its maximum value, while it decreases as the angle approaches 90°. This model represents a single forward lobe and does not include the backward hemisphere.

For three-dimensional visualization, the radiation pattern is converted from spherical coordinates to Cartesian coordinates using the following relations:

$$ X=R\sin\theta\cos\phi $$
$$ Y=R\sin\theta\sin\phi $$
$$ Z=R\cos\theta $$

where θ is the polar angle and ϕ is the azimuth angle.

The resulting 3D radiation pattern shows a concentrated beam along the positive Z-axis. This directional property makes horn antennas suitable for applications such as radar systems, satellite communication, microwave links, and antenna measurements. Horn antennas are also widely used as standard gain antennas because of their predictable radiation characteristics.

Thus, this experiment provides an understanding of the directional radiation characteristics of a horn antenna through a simplified mathematical model.

Source Code

```
# Horn Antenna - 3Dimport numpy as npimport matplotlib.pyplot as plt# Define anglestheta = np.linspace(0, np.pi/2, 250)phi = np.linspace(0, 2*np.pi, 250)theta, phi = np.meshgrid(theta, phi)# Define radiation patternn = 6R = np.cos(theta)**n# Convert to Cartesian coordinatesX = R * np.sin(theta) * np.cos(phi)Y = R * np.sin(theta) * np.sin(phi)Z = R * np.cos(theta)# Plot 3D radiation patternax = plt.figure().add_subplot(projection='3d')ax.plot_surface(X, Y, Z, cmap="inferno")ax.set_xlim(-1, 1)ax.set_ylim(-1, 1)ax.set_zlim(0, 1)ax.set_title("Horn Antenna (3D)")ax.set_xlabel("X")ax.set_ylabel("Y")ax.set_zlabel("Z")plt.show()
```

Output

Figure 3: 3D Radiation Pattern of Horn Antenna

Result and Discussion

The 3D radiation pattern of a horn antenna was successfully simulated using Python. The graph shows a dominant forward lobe along the positive Z-axis, indicating that the radiation is concentrated mainly in one direction. The radiation strength is maximum at $\theta=0^\circ$ and

decreases as the angle approaches \(90^\\circ\). The value \(n=6\) produces a relatively narrow beam, demonstrating the directional characteristics of the antenna. These characteristics make horn antennas suitable for radar systems, satellite communication, and microwave transmission. The simulation provides a simplified visualization of horn antenna radiation, although it does not represent the exact electromagnetic radiation pattern of a practical horn antenna.